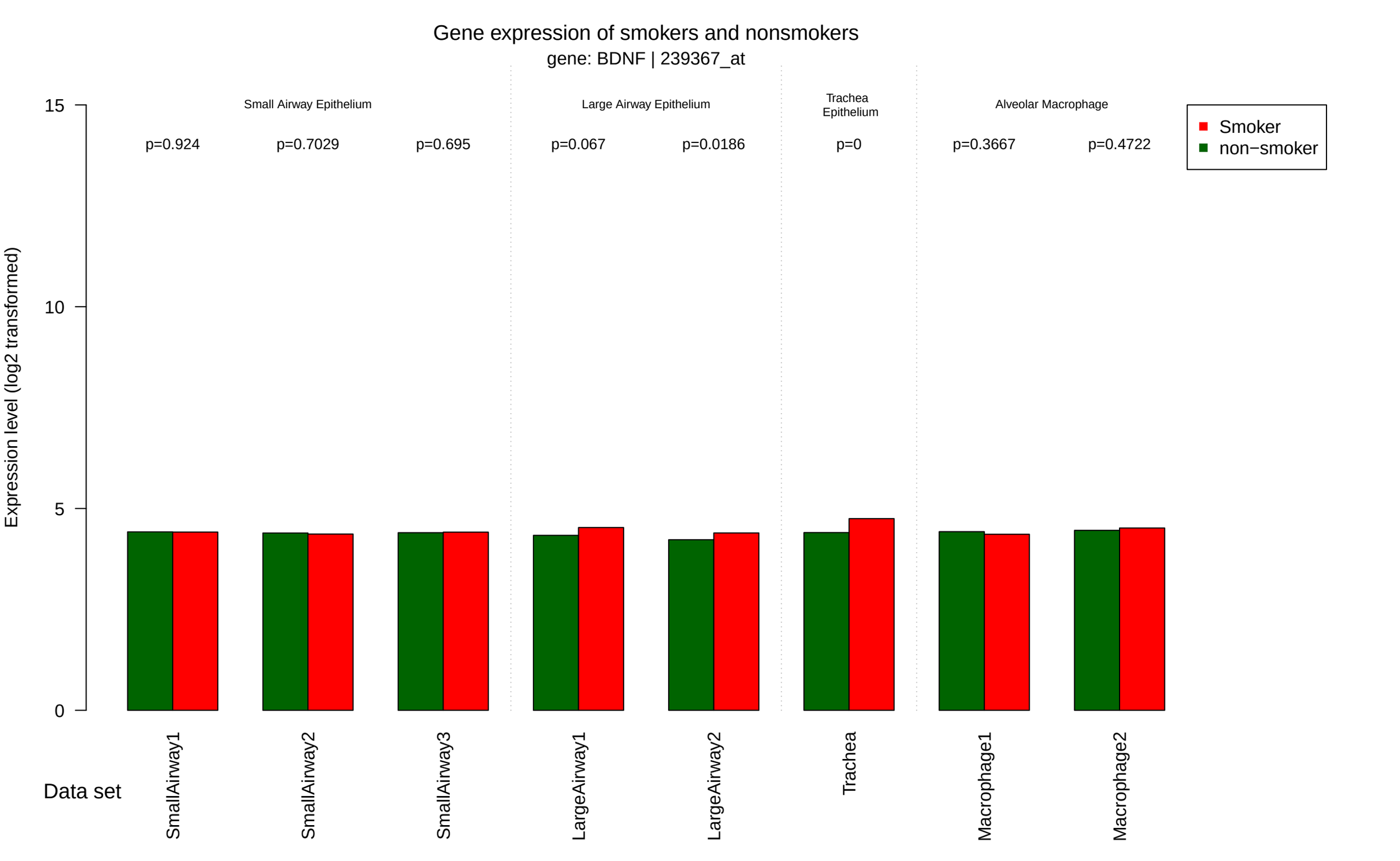
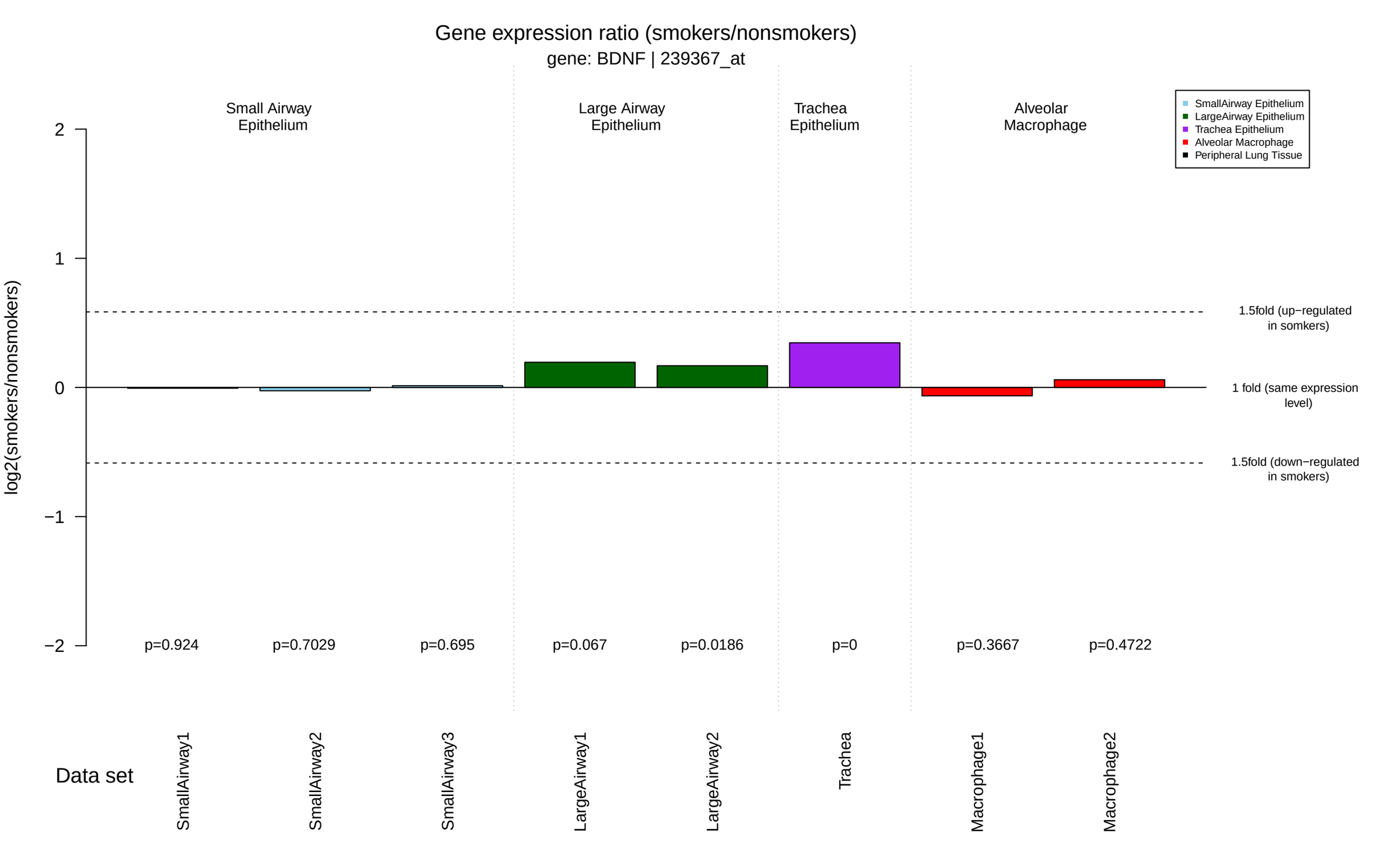
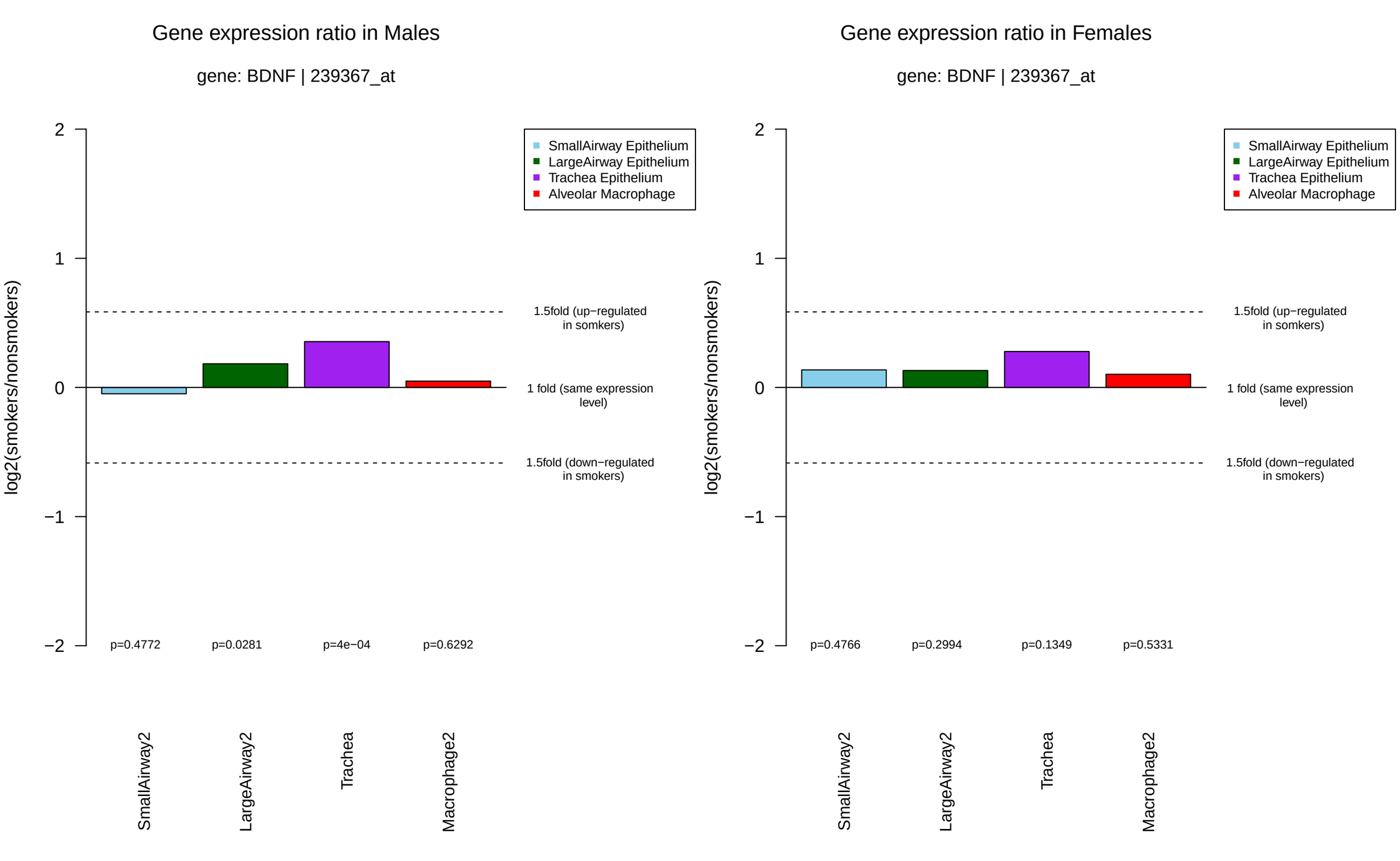
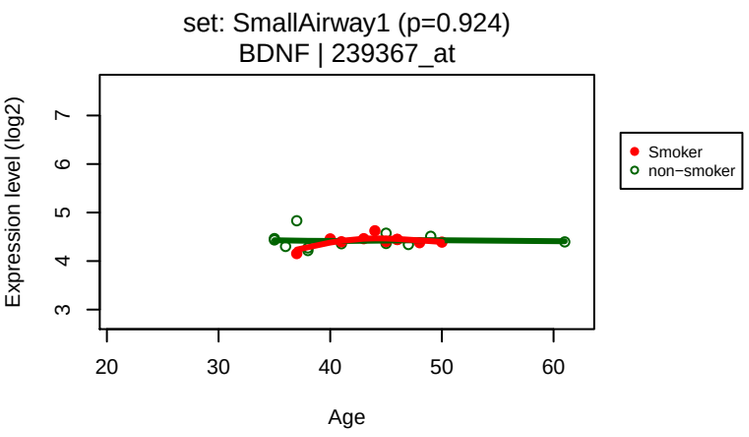
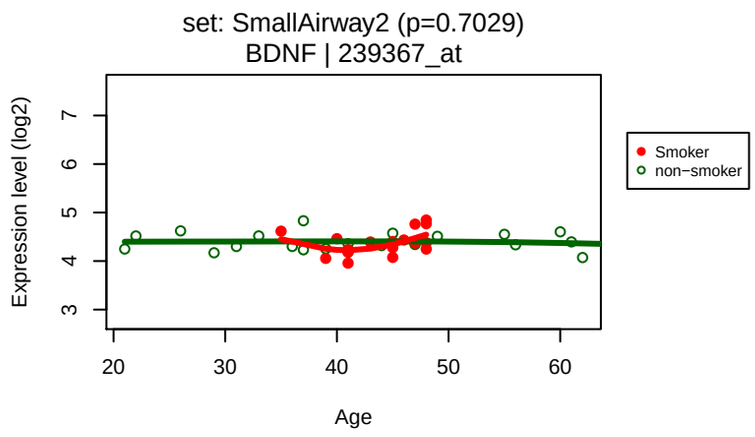
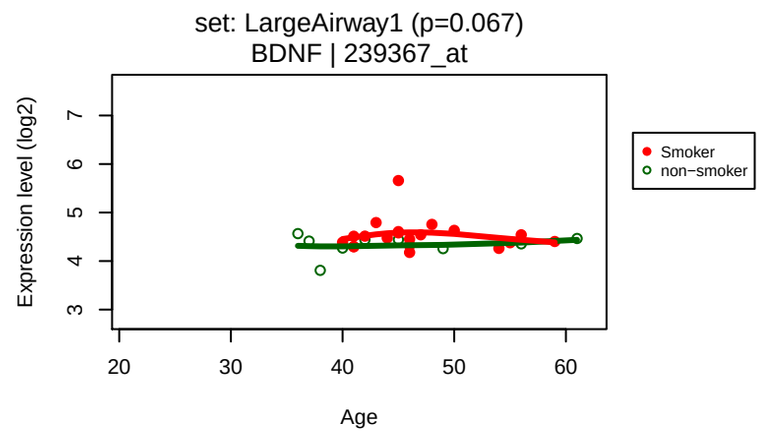

Gene expression of smokers and nonsmokers
gene: BDNF | 239367_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.924
p=0.7029
p=0.695
p=0.067
p=0.0186
p=0
p=0.3667
p=0.4722
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

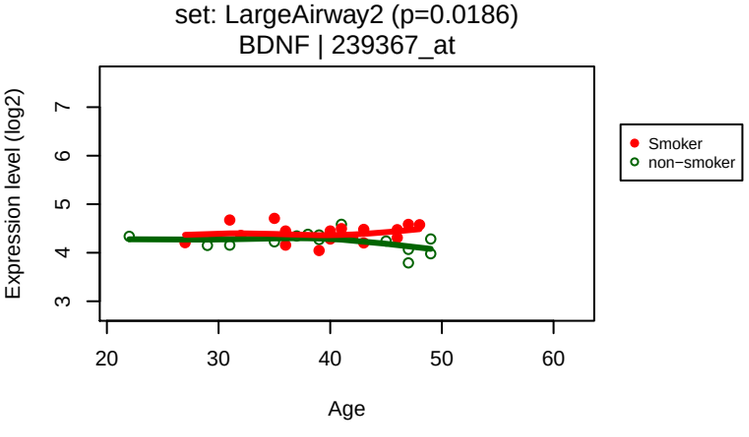

Gene expression ratio (smokers/nonsmokers)
gene: BDNF | 239367_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.924
p=0.7029
p=0.695
p=0.067
p=0.0186
p=0
p=0.3667
p=0.4722
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

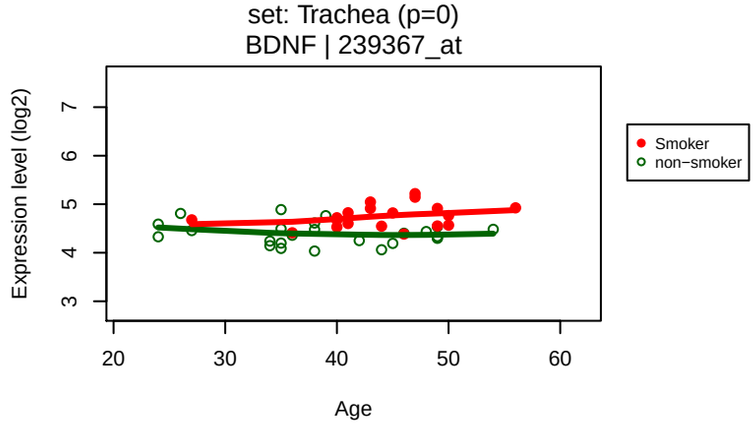

Gene expression ratio in Males
gene: BDNF | 239367_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4772
p=0.0281
p=4e−04
p=0.6292
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BDNF | 239367_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.4766
p=0.2994
p=0.1349
p=0.5331
SmallAirway2
LargeAirway2
Trachea
Macrophage2

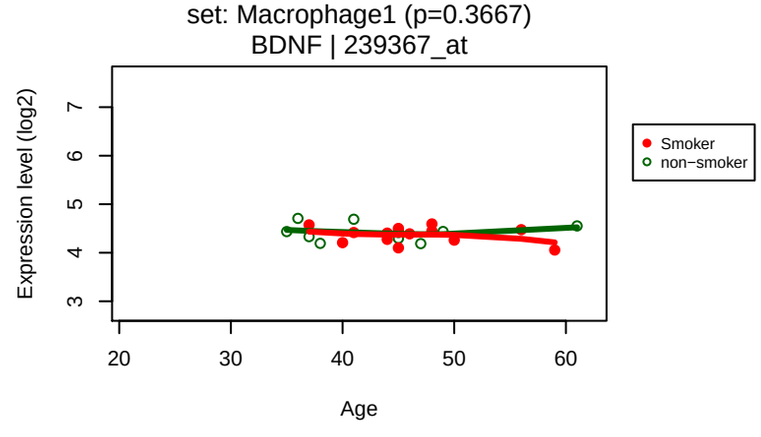

set: SmallAirway1 (p=0.924)
BDNF | 239367_at
Expression level (log2)
Age
Smoker
non−smoker

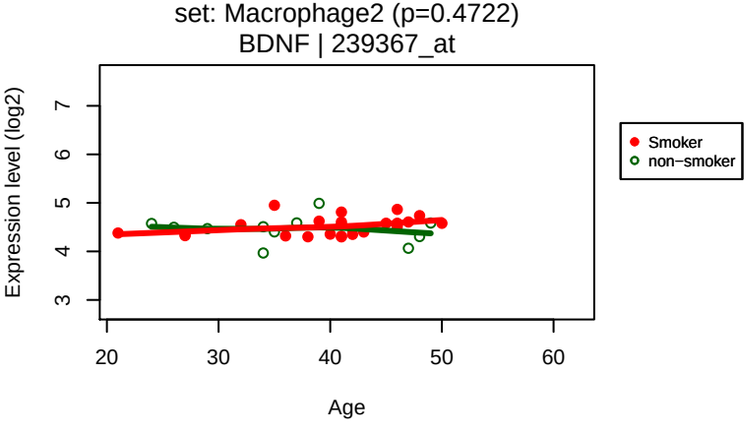

set: SmallAirway2 (p=0.7029)
BDNF | 239367_at
Expression level (log2)
Age
Smoker
non−smoker

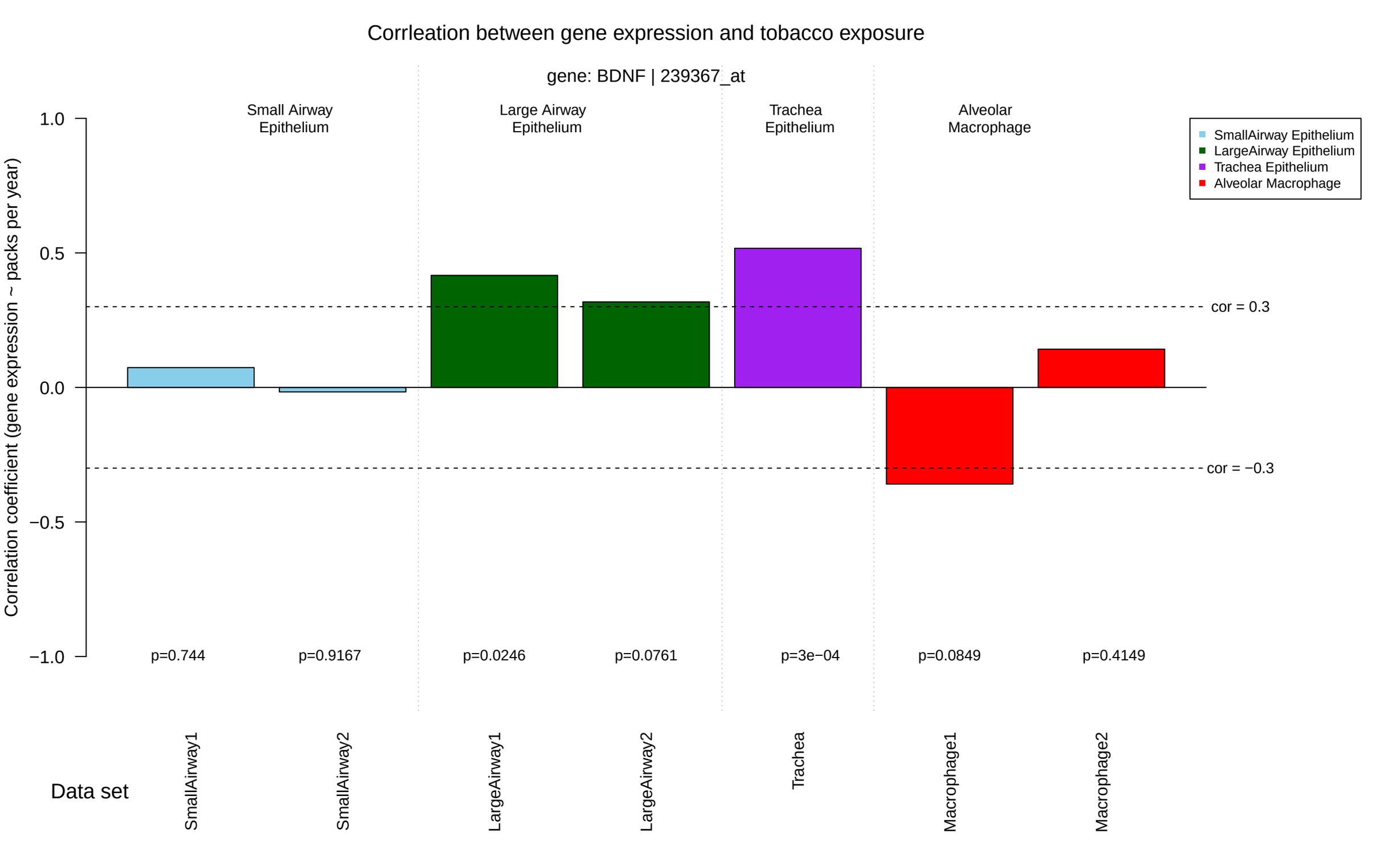

set: LargeAirway1 (p=0.067)
BDNF | 239367_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0186)
BDNF | 239367_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0)
BDNF | 239367_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3667)
BDNF | 239367_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.4722)
BDNF | 239367_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BDNF | 239367_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.744
p=0.9167
p=0.0246
p=0.0761
p=3e−04
p=0.0849
p=0.4149
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2